Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Name Surname

Automation QA Engineer
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Summary

Results-oriented and detail-driven Automation Quality Engineer with a strong background in Java programming and a passion for ensuring software reliability. Experienced in designing, implementing, and executing automated test scripts to verify the functionality and performance of software applications. Proven track record of collaborating with development teams to identify and resolve issues early in the development process. Adept at utilizing testing frameworks such as JUnit and incorporating continuous integration tools like Jenkins. Skilled in creating and maintaining automated test suites for web and mobile applications. Committed to delivering high-quality software by applying industry best practices in testing and automation. Seeking opportunities to contribute technical expertise and enhance software development processes as an Automation Quality Engineer.

1 Technical Skills

[leftmargin=0.15in, label=] **Languages**: Java, TypeScript, SQL**Frameworks**: Selenium/Selenide, Playwright, RestAssured, JUnit, TestNG, Cucumber**Developer Tools**: Git, Docker, Google Cloud Platform, Maven/Gradle, Grafana, Kibana, Splunk**Libraries**: WireMock, Spring Cloud Contract, Pact, Chaos Monkey

2 Experience

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Lead Automation QA Engineer Apr. 2020 – Feb. 2021
Somename-bank - it's a big bank with web, API and mobile applications. *City, Country*

- Created common testing framework (Java, Cucumber, RestAsssured, for component testing in 5+ teams
- Covered 100% manual scenarios with automation (Java, Selenide, Cucumber, RestAssured) that decreased regression time from 5 days to 2 days
- Integrated automation jobs into Jenkins and maintained test automation logs, reporting and verifying bug resolutions
- Successfully implemented new processes across 8 teams: created test quality metrics, training materials, organized onboarding, improving work processes for production defects
- Contributed to candidate interviews and mentored QA engineers, providing guidance and assigning tasks
- Awarded as "Engineer of the Year - QA", "Delivery team of the Year"

- **Senior Quality Assurance Engineer** Jul 2017 - Feb 2018
  Experio, OpenBank *City, Country*
  - Manual testing for a big bank. Worked with CFT, MS Dynamics CRM, 3Card-R, OpenWay Way4.

- **QA Engineer** Feb 2014 - Jul 2017
  LANIT BPM *City, Country*
  - Project with medical desktop application, where I participated as a manual QA Engineer and mentored 3 juniors.

3 Education

[leftmargin=0.15in, label=]
Name State Technical University City, Country
Engineer's Degree, Faculty of Electronics and Instrument Making *2007 - 2012*

4 Certifications

[leftmargin=0.15in, label=] **Cloud Digital Leader - Google Cloud***Issued Sep 2022 - Expires Sep 2025*